

Thermal Second-Order Response and Sharper Chain-Transfer Bounds for the MQM Latch

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Abstract—We execute full covariance-weighted two-excitation response calculations for finite Legendre-chain references of 32, 64, 128 and 192 modes. At 192 modes, the simulated retained second-order half trace norm is approximately 2.2506×10^{-8} . An alternative Chebyshev transfer derivation reduces the sufficient vacuum-chain length from 3072 to 1536 sites at the same 2^{-128} transfer budget. A separate exact-arithmetic covariance comparison for 192 Gauss nodes yields a response allowance below 2^{-57} . These results concern distinct reference sizes: the 1536-site vacuum theorem does not certify the 192-mode propagation. Directed numerical and vacuum transfer errors at the evaluated size remain open, so the original finite-temperature 2^{-16} target is not promoted.

Index Terms—Quantum control, Gaussian reservoirs, second-order response, numerical certification.

I. FROZEN SOURCE AND REFERENCE SEPARATION

The source contract is $\Omega = 64$, $B = 16$, $\kappa = 8$, $\beta = 5/16$, $T = 16$, $h = g = 1$, $\lambda^2 = 4\sqrt{2}$, and fixed native launch $p0$ with reservoir vacuum in the response representation. The native eight-state register has charges $(1, 1, 1, 1, 0, 2, 2, 2)$ in order $(p0, m0, e0, d1, d0, p1, m1, e1)$. The local drive operator is $S = I_C \otimes \sigma_R^-$. Untouched banks and references are carried identically; this is a fixed-launch response certificate, not a general input diamond-norm theorem.

The exact Gaussian continuum covariance is

$$C(s) = \frac{\kappa}{2\pi} \int_{-B}^B n_\beta(\Omega + \delta) e^{-i\delta s} d\delta. \quad (1)$$

The inherited envelope is $T^2V < 16384/31457265$, where $V = C(0)$. The orthogonal-polynomial chain mapping is established prior art [1]. Flat coupling maps to site zero, with $G^2 = 128/\pi$ and links $b_n = Bn/\sqrt{4n^2 - 1}$.

We distinguish two models. The transfer theorem truncates *vacuum propagation only* and retains exact $C(s)$. The numerical model is a finite stationary Legendre chain with $N = f(J_M)$, where $f(x) = n_\beta(\Omega + x)$; it replaces both vacuum propagation and covariance. The second replacement is separately bounded below. Neither model supplies a physical sink, material implementation, thermal refresh or hardware evidence.

II. FULL-COLUMN FINITE RESPONSE

Let (x_j, w_j) denote Gauss–Legendre nodes and weights on $[-1, 1]$. The finite Jacobi matrix has these spectral coordinates. Its star representation has

$$\delta_j = 16x_j, \quad c_j^2 = 64w_j/\pi, \quad N_{jj} = n_\beta(64 + 16x_j). \quad (2)$$

MQM development release v0.1, 7 October 2026. Conditional derivations and constructor controls. Independent review pending; physical promotion zero.

TABLE I
SIMULATED FINITE STATIONARY-CHAIN RESPONSE; BINARY64.

M	D_2	$\frac{1}{2} \ \Delta_2^M\ _1$	Cold live probability
32	659	9.2882×10^{-8}	0.63853
64	2339	5.0426×10^{-8}	0.33841
128	8771	2.2774×10^{-8}	1.6313×10^{-5}
192	19299	2.2506×10^{-8}	1.0713×10^{-5}

The covariance is $C_M(s) = \sum_j c_j^2 N_{jj} e^{-i\delta_j s}$. The total excitation sectors are invariant; $D_1 = M + 4$ and

$$D_2 = 3 + 4M + M(M + 1)/2. \quad (3)$$

For this finite stationary passive Gaussian bath, the source response identity [2] is

$$\Delta_2^M = \sum_{j=1}^M N_{jj} \rho_j(T) - (\text{tr } N) \rho_{\text{vac}}(T), \quad (4)$$

where ρ_j is the retained output from $p0$ with one initial photon in mode j . It is a Gaussian derivative identity, not truncation of the full thermal state. All M columns are propagated, not a selected one-hot pilot. The two terms are combined at finite M ; no separate divergent $\text{tr } N$ limit is taken.

The exponential-action implementation uses sparse matrices and full source columns. An independent $M = 4$ occupation-basis dense constructor agrees exactly in its binary64 matrix entries and within 2.61×10^{-20} in retained hot output. A two-half-step $M = 32$ comparison differs by 2.53×10^{-19} . Trace and Hermiticity diagnostics pass the frozen 10^{-12} controls. None is a directed error certificate. The $M = 192$ matrix has 95236 nonzeros and 192 source columns.

The cold diagnostic initially counted auxiliary $d1$ as captured. The recorded amendment counts only $d0$ as captured; the computed response matrices are unchanged. The original receipt is retained. The $M = 192$ cold diagnostic agrees numerically with the inherited continuum reference, but agreement is not a continuum enclosure.

III. ALTERNATIVE VACUUM TRANSFER AUDIT

Write H_∞ for the infinite vacuum chain and H_L for its cut after site $L-1$. Both have norm at most $\Lambda = 64$ on $Q \leq 2$. The source paired double-commutator response consists of eight covariance-weighted dyads. Each ket or bra has at most three propagation segments, with total duration T , and at most two local insertions. Words of total Hamiltonian degree less than L coincide: the local port and nearest-neighbor bath cannot reach an omitted site in fewer factors. This is the same launch and support contract as the parent Taylor proof [3].

Theorem 1. *Keeping the exact continuum covariance in both models, define $x = \Lambda T/2$. Then*

$$\frac{1}{2} \|\Delta_2^\infty - \Delta_{2,L}^C\|_1 \leq 64T^2V \sum_{n=L}^{\infty} \frac{x^n}{n!}. \quad (5)$$

Proof. The Chebyshev expansion of each propagator has coefficients bounded by $2|J_n(\Lambda t)|$. The Bessel Poisson integral [4] gives $|J_n(a)| \leq (|a|/2)^n/n!$ for real a . Multiplying at most three expansions and grouping total degree gives a coefficient majorant $8x^n/n!$. Each Chebyshev polynomial is a contraction and has degree n . Lower-degree words agree. Two model tails therefore give ket difference at most $16 \sum_{n \geq L} x^n/n!$. Normalized dyads differ by at most twice that amount. Eight dyads, the half trace factor and triangle area $T^2/2$ give (5). This same-agent alternative proof is not independent external review. $\square$

At $x = 512$ and $L = 1536 = 3 \cdot 512$, exact integer comparison gives

$$\frac{1048576}{31457265} \frac{512^{1536}}{1536! [1 - 512/1537]} < 2^{-229}. \quad (6)$$

This clears the frozen 2^{-128} budget with half the prior sufficient site count. The $Q = 2$ reference dimension falls from 4732419 to 1186563. This is a mathematical reference reduction, not physical qubit or comparative advantage evidence.

IV. SEPARATE THERMAL COVARIANCE COMPARISON

Gauss quadrature [4] integrates polynomials through degree $2M - 1$ exactly. Expand the Bose factor as a positive exponential sum and set $r = e^{-15} < 2^{-21}$. For Bose terms $1 \leq \ell \leq 8$ and $|s| \leq 16$, the complex Chebyshev argument obeys

$$\frac{B}{2} \sqrt{s^2 + (\beta\ell)^2} \leq 128 + 25\ell^2/1024 \leq 2073/16 =: y. \quad (7)$$

The complex Bessel Poisson bound [4] includes $e^{B\beta\ell}$, combining it with $e^{-\beta\Omega\ell}$ leaves r^ℓ . Quadrature and integral each have unit mass in the normalized flat measure. Thus, with $G^2 < 128/3$, an exact rational majorant for $\eta_M = \sup_{|s| \leq T} |C(s) - C_M(s)|$ is

$$\eta_M \leq 4G^2 \frac{r}{1-r} \frac{y^{2M}}{(2M)! [1 - y/(2M+1)]} + 2G^2 \frac{r^9}{1-r}. \quad (8)$$

The final term bounds the remaining Bose sum directly. For response at fixed vacuum propagator, eight unit-norm dyads give covariance replacement debit at most $2T^2\eta_M$. Substituting $M = 192$ and rational envelopes in (8) gives an exact comparison below 2^{-57} . This statement concerns ideal Gauss nodes and weights; binary64 implementation differences still belong to the numerical ledger.

V. UNCLOSED GATE AND NEXT WORK

For a numerical stationary-chain response, the remaining sufficient gate is

$$\frac{1}{2} \|\hat{\Delta}_2^M\|_1 + \delta_{\text{vac}}(M) + 2T^2\eta_M + \delta_{\text{num}} < 127/2^{25} - 2^{-128}. \quad (9)$$

The 192-mode estimate is encouraging, but no adequate $\delta_{\text{vac}}(192)$ or directed δ_{num} is supplied. The 1536-site vacuum theorem cannot be attached to the 192-mode numerical result. Conversely, the 192-node covariance theorem does not close vacuum propagation. The analytic covariance comparison and dense control were added after numerical evaluation began, with this development amendment disclosed; source parameters and targets were unchanged. Next work must certify vacuum observable transfer at the evaluated size or execute the response at a certified size, then enclose numerical error. The primary thermal target remains OPEN; physical promotion remains zero.

REFERENCES

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